Dynamic Modeling of Elevator System with fault diagnosis

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1 Introduction

Elevator systems are widely used in modern buildings and their performance is strongly influenced by vibration and dynamic behaviour. Mechanical vibrations in elevators can arise due to irregular acceleration profiles, rail misalignment, traction system disturbances, or structural imperfections. These vibrations not only affect passenger comfort but may also indicate developing mechanical faults.

To analyse the vibration behaviour of the elevator system, a dynamic model is developed considering the elevator car, counterweight, and traction sheave. The system is represented as a three degree-of-freedom mechanical system consisting of masses, damping elements, and stiffness elements. The motion of the system is described using displacement variables corresponding to each component.

The system variables are defined as follows:

- x_1 : Vibration line displacement of the elevator car
- x_2 : Vibration line displacement of the counterweight
- x_3 : Vibration line angular displacement of the traction sheave

counterweight, and traction sheave dynamics.

The dynamic behavior of the elevator system can be represented using the standard second-order mechanical system equation:

$$M\ddot{x} + C\dot{x} + Kx = F \tag{1}$$

where M, C, and K represent the mass, damping, and stiffness matrices of the system, respectively.

The mass matrix is defined as

$$M = \begin{bmatrix} m_c & 0 & 0 \\ 0 & m_w & 0 \\ 0 & 0 & J_{Tr} \end{bmatrix} \tag{2}$$

where m_c is the mass of the elevator car, m_w is the mass of the counterweight, and J_{Tr} is the moment of inertia of the traction sheave.

The damping matrix of the system is given by

$$C = \begin{bmatrix} c_c + c_g & 0 & c_c R_{Tr} \\ 0 & c_w + c_g & -c_w R_{Tr} \\ c_c R_{Tr} & -c_w R_{Tr} & (c_c + c_w) R_{Tr}^2 \end{bmatrix} \tag{3}$$

The stiffness matrix is expressed as

$$K = \begin{bmatrix} k_c + k_g & 0 & k_c R_{Tr} \\ 0 & k_w + k_g & -k_w R_{Tr} \\ k_c R_{Tr} & -k_w R_{Tr} & (k_c + k_w) R_{Tr}^2 \end{bmatrix} \tag{4}$$

In addition to the primary damping and stiffness parameters, guide rail damping (c_g) and guide rail stiffness (k_g) are included in the model. These parameters represent the interaction between the elevator car or counterweight and the guide rails along which they move.

The guide rails provide structural support and help maintain vertical alignment of the elevator system. However, friction and mechanical contact between the guide shoes and rails introduce additional damping and stiffness effects. These effects influence the vibration characteristics of the elevator system and therefore must be considered in the dynamic model.

Including c_g and k_g allows the simulation model to more accurately capture vibration behaviour caused by rail contact, misalignment, and structural flexibility.

Elevator Car Equation

$$m_c \ddot{x}_1 + (c_c + c_g)\dot{x}_1 + c_c R_{Tr} \dot{x}_3 + (k_c + k_g) x_1 + k_c R_{Tr} x_3 = m_c a \tag{5}$$

Counterweight Equation

$$m_w \ddot{x}_2 + (c_w + c_g)\dot{x}_2 - c_w R_{Tr} \dot{x}_3 + (k_w + k_g) x_2 - k_w R_{Tr} x_3 = m_w a \tag{6}$$

Traction Sheave Equation

$$J_{Tr} \ddot{x}_3 + c_c R_{Tr} \dot{x}_1 - c_w R_{Tr} \dot{x}_2 + (c_c + c_w) R_{Tr}^2 \dot{x}_3 + k_c R_{Tr} x_1 - k_w R_{Tr} x_2 + (k_c + k_w) R_{Tr}^2 x_3 = \frac{2 J_{Tr}}{R_{Tr}} a \tag{7}$$

where

- m_c : mass of elevator car
- m_w : mass of counterweight
- J_{Tr} : moment of inertia of traction sheave
- R_{Tr} : radius of traction sheave

- c_c, c_w : damping coefficients
- c_g : guide rail damping
- k_c, k_w : stiffness coefficients
- k_g : guide rail stiffness
- a : elevator acceleration input

2 Concepts Implemented in the Model

The developed mathematical model and simulation framework incorporate several key concepts required for realistic modeling of elevator vibration dynamics. The implemented concepts are described below.

2.1 Multi-Degree-of-Freedom Dynamic Modeling

The elevator system is modeled as a second-order multi-degree-of-freedom (MDOF) system using the equation:

$$M\ddot{x} + C\dot{x} + Kx = F \tag{8}$$

This formulation captures the coupled vibration behavior of the elevator car, counterweight, and traction sheave. The interaction between these components is represented through damping and stiffness coupling terms in the system matrices.

2.2 Motor-to-Car Vibration Transmission

The model captures the transmission of vibrations from the motor side to the elevator car through intermediate components such as the traction sheave and rope system. This is achieved through:

- Traction sheave rotational dynamics (J_{Tr})
- Rope-pulley coupling via traction radius (R_{Tr})
- Counterweight interaction
- Guide rail effects (c_g, k_g)

These elements ensure that vibrations propagate through the system, allowing realistic modeling of how disturbances affect the passenger cabin.

2.3 Multi-Source Excitation Modeling

Instead of a single input, the system incorporates multiple excitation sources to represent real-world operating conditions. The total input acceleration is defined as:

$$a(t) = a_{ref}(t) + a_{motor}(t) + a_{cw}(t) + a_{guide}(t) + a_{pulley}(t) + n(t) \tag{9}$$

This includes contributions from motor ripple, counterweight motion, pulley dynamics, guide rail interaction, and noise.

2.4 Superposition Principle Implementation

The principle of superposition is implemented by combining all excitation sources using a summation block in the simulation model. The resulting input signal represents the total vibration acting on the system:

$$a_{total}(t) = \sum_{i=1}^{n} a_i(t) \tag{10}$$

This ensures that the overall vibration response is a combination of all individual sources, leading to realistic waveform generation.

2.5 Frequency Domain Analysis Capability

The model supports frequency-domain analysis using the Fast Fourier Transform (FFT). The generated vibration signals can be decomposed into their constituent frequency components, allowing identification of dominant vibration sources.

Distinct frequency peaks corresponding to motor ripple, counterweight motion, and pulley dynamics can be observed, validating the effectiveness of the model.

2.6 Model Extensibility for Fault Simulation

The developed framework is designed to support future implementation of fault conditions. By modifying input signals and system parameters such as stiffness, damping, and disturbance levels, different fault scenarios can be simulated.

This flexibility makes the model suitable for future extension toward fault diagnosis and predictive maintenance applications.

2.7 Inputs Applied to the Model

Based on the developed excitation model, multiple vibration sources are applied to the system through a combined acceleration input. These inputs are implemented in the simulation using summation blocks and represent realistic operating conditions of the elevator.

The total input acceleration applied to the system is given by:

$$a(t) = a_{ref}(t) + a_{motor}(t) + a_{cw}(t) + a_{guide}(t) + a_{pulley}(t) + n(t) \tag{11}$$

The individual components of the input are defined as follows:

- $a_{ref}(t)$: Smooth acceleration profile representing normal elevator motion
- $a_{motor}(t)$: Motor-induced ripple vibrations (20 Hz and 40 Hz components)
- $a_{cw}(t)$: Counterweight-induced vibration (low frequency, $\sim$5 Hz)
- $a_{guide}(t)$: Guide rail interaction vibration (medium frequency, $\sim$8 Hz)
- $a_{pulley}(t)$: Pulley/sheave vibration (higher frequency, $\sim$12 Hz)
- $n(t)$: Small random noise representing measurement disturbances

These inputs are combined using the principle of superposition and applied to the dynamic system to generate realistic vibration responses.

2.8 Vibration Response Results

The response of the elevator system is analyzed in terms of displacement, velocity, and acceleration of the elevator car. The primary variable of interest is x_1, which represents the vibration displacement of the elevator car.

The corresponding velocity and acceleration responses are obtained as $\dot{x}_1$ and $\ddot{x}_1$, respectively. These signals provide insight into the dynamic and vibrational characteristics of the system.

In addition to the individual responses, a combined vibration signal is considered to represent the total vibration experienced inside the elevator car.

The results show that the displacement response remains smooth, while the velocity and acceleration responses capture the dynamic variations introduced by multiple vibration sources. The combined signal a_{total} reflects the superposed effect of all excitation components, providing a realistic representation of elevator vibration behaviour.

The Simulink model represents the three degree-of-freedom dynamic system of the elevator, consisting of the elevator car (x_1), counterweight (x_2), and traction sheave (x_3). The model is structured using three subsystems, each implementing the corresponding second-order differential equation derived from the mathematical model.

The input to the system is a combined acceleration signal generated using a summation block, where multiple vibration sources such as the reference motion, motor ripple, counterweight effects, pulley vibrations, guide rail interactions, and noise are superimposed. This combined input is applied to each subsystem in the form of force.

Inside each subsystem, the net force is computed using gain and summation blocks representing mass, damping, and stiffness effects. The resulting acceleration is obtained by multiplying with the inverse mass/inertia term, and two integrator blocks are used to compute velocity and displacement.

The outputs of the model include displacement, velocity, and acceleration of each component, which are exported to the workspace for further vibration analysis. The modular structure of the model allows clear representation of system dynamics and easy modification of parameters.

In addition to the system outputs, scope blocks are used in the Simulink model to visualize the input signals and their combined effect. A multi-input scope is used to display all individual excitation components, including the reference acceleration, motor ripple, counterweight vibration, guide rail disturbance, pulley vibration, and noise signal. This helps in verifying the presence and contribution of each vibration source.

Another scope is used to display the output of the summation block, which represents the combined acceleration signal obtained after applying the principle of superposition. This signal serves as the effective input to the dynamic system. The visualization of both individual inputs and the combined signal provides clarity on how multiple vibration sources interact to produce the overall excitation applied to the elevator system.

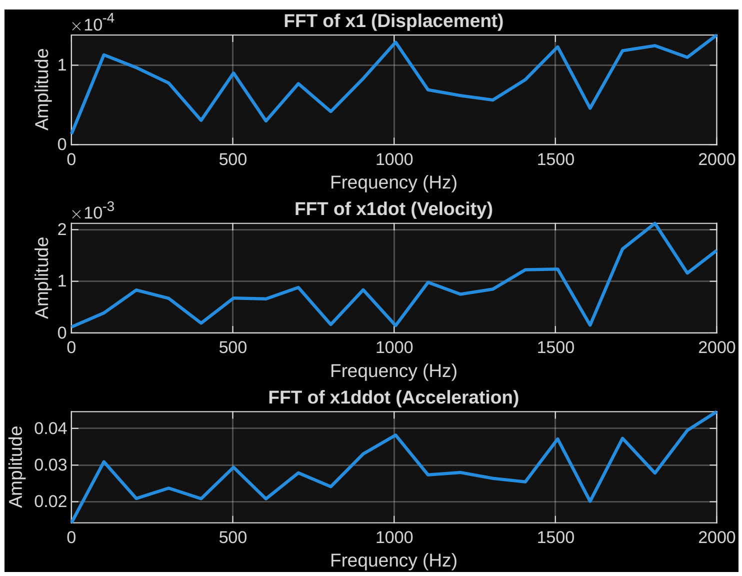

Figure 1: Elevator car Vibrations in Frequency Domain

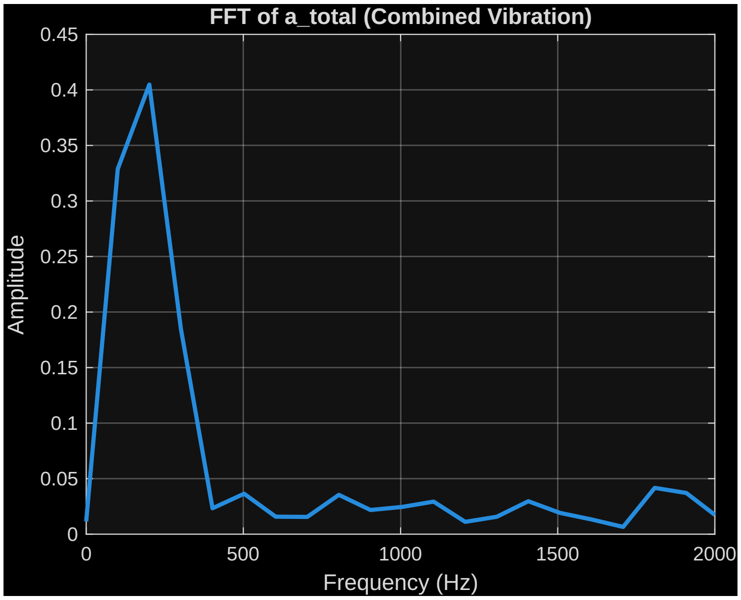

Figure 2: Input given to elevator car in Frequency domain

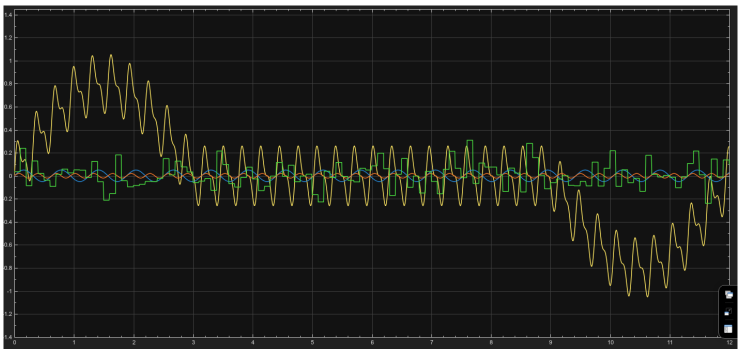

Figure 3: All inputs given to the Elevator car

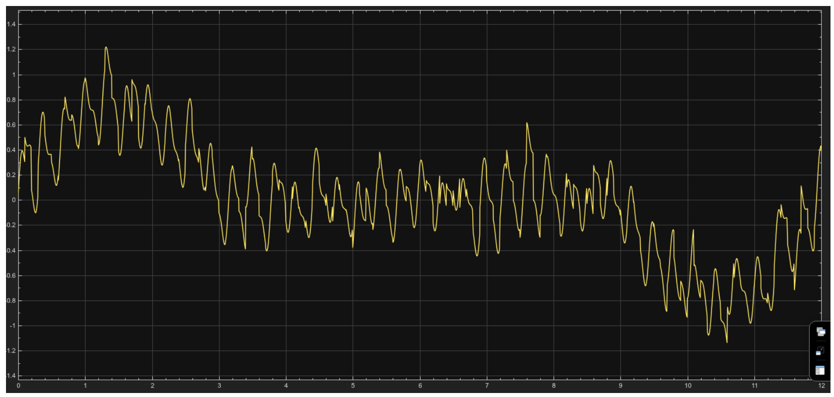

Figure 4: Output shown after superposition principle for all given inputs

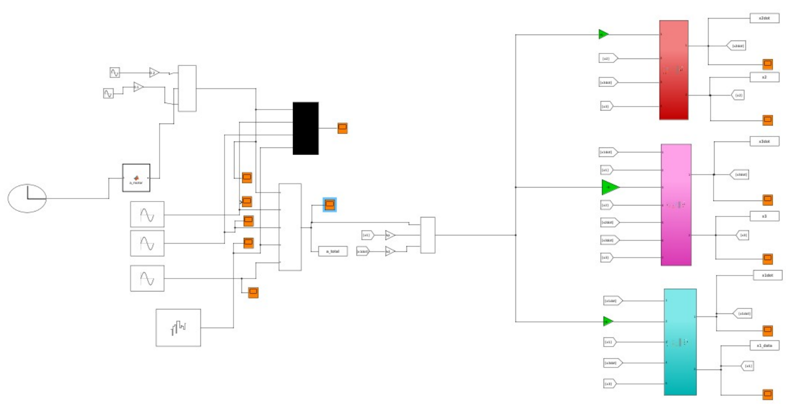

Figure 5: Simulink model of elevator

The model therefore provides a strong foundation for further extension toward fault simulation and predictive maintenance applications.

3 Future Scope: Data Generation for Fault Diagnosis

The developed mathematical model provides a strong foundation for generating synthetic vibration datasets for elevator fault diagnosis and predictive maintenance applications. In future work, different fault conditions can be simulated by modifying system parameters and input excitation profiles.

For each operating condition, the model can generate time-domain vibration signals including displacement (x_1), velocity ($\dot{x}_1$), acceleration ($\ddot{x} * 1$), and the combined vibration signal ($a*total$). These signals can be segmented into fixed-length samples to form a structured dataset.

Different fault scenarios can be introduced as follows:

- **Normal Condition:** Smooth acceleration input with nominal system parameters, representing healthy operation.
- **Start–Stop Fault:** Oscillatory disturbances added to the acceleration profile during steady motion.
- **Jump Fault:** Sudden spikes introduced in the acceleration input to simulate mechanical shocks or jerks.
- **Acuteness (Rough Ride):** High-frequency oscillations added to represent structural or stiffness-related issues.

For each condition, multiple samples can be generated by varying parameters such as noise level, vibration amplitude, and system properties. The resulting dataset can be labeled according to the operating condition and used to train machine learning models.

Additionally, frequency-domain features can be extracted using Fast Fourier Transform (FFT), enabling the identification of characteristic frequency signatures for each fault type. This allows the dataset to be used for classification, fault detection, and predictive maintenance of elevator systems.

Thus, the proposed model serves as a scalable platform for generating realistic and diverse vibration datasets suitable for intelligent fault diagnosis applications.